

GCM-SST in QUIC



[draft-westerlund-tls-gcm-sst-00](#)

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GCM-SST Introduction

- Galois Counter Mode with Strong Secure Tags (GCM-SST) is an AEAD Algorithm
 - [draft-mattsson-cfrg-aes-gcm-sst-21](#)
 - Standardized by ETSI / 3GPP
 - Two independent security proofs
 - Uses the same cryptographic constructs as AES-GCM
 - Modifies the Authentication hash construction
 - Derives a unique key for each packet protected
 - Fixes a known weakness preventing truncation of AES-GCM Authentication tags
- GCM-SST is defined for three Algorithms:
 - AES-128 with 128-bit keys and 128-bit blocks
 - AES-256 with 256-bit keys and 128-bit blocks
 - Rijndael with 256-bit keys and 256-bit blocks

Name	Tag length (bytes)	Forgery probability before first forgery	Forgery probability after first forgery	Expected number of forgeries
GCM_SST_14	14	$1 / 2^{112}$	$1 / 2^{112}$	$v / 2^{112}$
GCM_SST_12	12	$1 / 2^{96}$	$1 / 2^{96}$	$v / 2^{96}$
POLY1305	16	$1 / 2^{91}$	$1 / 2^{91}$	$v / 2^{91}$
GCM	16	$1 / 2^{116}$	1	$v^2 / 2^{117}$

v: Number of forgery attempts

Values assuming packets up to 65536 bytes

QUIC usage of GSM-SST

- Draft proposes 96 bit (12 bytes) and 112 bit (14 bytes) authentication tags for the three configurations
 - 96-bit auth tags are 5-bit stronger than ChaCha20-POLY1305 with 4-byte shorter tags.
- Uses AES-ECB (same as AES-GCM) for header protection
 - Rijndael needs sample extension for the nonce:
mask = Stream(40, hp_key, ZeroPad(sample, 28))[0..4]

Cipher Suite Name	AEAD Algorithm	Hash Algorithm	Tag Length (bits)
TLS_AES_128_GCM_SST_12_SHA256	AEAD_AES_128_GCM_SST_12	SHA256	96
TLS_AES_128_GCM_SST_14_SHA256	AEAD_AES_128_GCM_SST_14	SHA256	112
TLS_AES_256_GCM_SST_12_SHA384	AEAD_AES_256_GCM_SST_12	SHA384	96
TLS_AES_256_GCM_SST_14_SHA384	AEAD_AES_256_GCM_SST_14	SHA384	112
TLS_RIJNDAEL_GCM_SST_12_SHA384	AEAD_RIJNDAEL_GCM_SST_12	SHA384	96
TLS_RIJNDAEL_GCM_SST_14_SHA384	AEAD_RIJNDAEL_GCM_SST_14	SHA384	112

Benefits in QUIC usage

- The Truncated Authentication Tags behaves like ideal authentication tags for unicast usage:
 - Forgery probability is approximately $1 / 2^t$ where t is tag length in bits.
- Even if attacker gets lucky with a forgery
 - AES-GCM: Subsequent forgeries until rekey are trivial
 - GCM-SST does not collapse, same difficult to do forgery on next packet.
- Saves 4 bytes per packet compared to AES-GCM and ChaCha20-Poly1305
- Rijndael enables 256-bit block crypto
 - Maximum number of decryption invocation much larger 2^{118} (instead of 2^{54})
 - Other limits are also larger but with limited utility in QUIC

Benefits for Multi-path and Multicast

- GCM-SST Multicast / Broadcast properties are improved
 - This is relevant for the Multicast proposals
 - Authentication does not behave ideal in this usage as the receivers / path are not independent and uniquely keyed
 - A successful forgery on one receiver reveals the auth key for that nonce: Use key to create forgeries for other receivers
 - Forgery attempts can be spread among the R receivers with unique packets
 - Assuming replay protection the expected number of distinct forgeries are $\approx v \cdot r / 2^t$
- Multipath
 - AES-GCM
 - The same Authentication key H is used for all paths just different nonces
 - Attacks can be distributed over all paths
 - GCM-SST is still ideal in multipath as each packet uses unique key
 - Attack must target specific packet (Nonce) on specific path

Interest

- Knowing if interest exist is useful for CFRG
- What authentication tag lengths are of interest?



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